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Programme Committee
ISMB 2026

Dear Programme Committee Members,

I am pleased to submit the manuscript entitled **"FACEO-R: Functionally Anchored Overlapping Community Detection for Robust Pathway Module Discovery"** for consideration at ISMB 2026.

Summary. Partitioning the genes of a pathway into functional modules is a routine step in transcriptomic analysis, and the field relies on three implicit assumptions: that a single partitioning algorithm is adequate, that a single resolution is defensible, and that the modularity Q of the resulting modules evidences biological organisation. None of these assumptions has been systematically tested within pathway subnetworks. We assembled a paired benchmark of 16 expression cohorts, nine GO biological-process pathways and 31 partitioning algorithms drawn from eight methodological families, with all algorithms operating on an identical topological overlap matrix. Each algorithm–pathway–cohort triple was evaluated across six orthogonal layers (structure, functional coherence, stability, robustness, biological validity and downstream clinical prediction), complemented by synthetic benchmarks with known ground truth.

Principal findings. Multi-resolution community detection produced functionally more coherent modules than hierarchical clustering (median $z = 0.689$ versus 0.045; paired Wilcoxon $q = 4.6 \times 10^{-7}$) and recovered nested functional sub-structures in synthetic benchmarks where hierarchical cutting merged them (ARI 1.000 versus 0.908). Downstream clinical prediction concurred (AUC 0.989 versus 0.898). Motivated by the observation that expression graphs degrade under realistic constraints, we further introduce **FACEO-R**, a functionally anchored overlapping community detection method in which the weight of GO-derived functional similarity adapts to the estimated reliability of the expression graph. On controlled degradation (sample subsampling, noise and feature dropout), FACEO-R outperformed Leiden $\gamma = 1.0$ in 124/165 comparisons (75%; $p = 9.2 \times 10^{-12}$) and WGCNA-style hierarchical cutting in 136/165 (82%; $p = 4.9 \times 10^{-19}$). The advantage replicated on *real* single-cell data (GSE145926; 69% and 94% win rates; $p = 0.029$ and 6.1×10^{-5}).

Why this is of interest to ISMB. Beyond the algorithmic comparison, our study delivers four methodological results that we believe are of general relevance to the computational biology community:

1. **Modularity is unstable with respect to graph construction.** Varying the edge-weight threshold from the 50th to the 95th percentile moves mean modularity from 0.269 to 0.419 (+56%) and *reverses* the sign of the comparison against a degree-preserving null model; weights contribute 29% of Q independently of topology. Modularity and its apparent significance should therefore not be reported as primary evidence.
2. **The functionally optimal resolution is not predictable.** Optimal γ was highly dispersed across pathways (entropy 1.74 of a maximum 1.79), and four families of automatic selection strategies all

failed (accuracy 0.16–0.46, none significantly above chance). Reporting a single arbitrary resolution is not defensible.

3. **We supply a remedy.** FACEO-R is never worse than standard Leiden on clean data (it collapses to it as graph reliability approaches unity) and is reliably better when expression data are limiting with an independent validation on real single-cell data.
4. **Enrichment p-values are size-biased.** We propose a size-matched, pathway-internal null that removes both the size and background biases.

We additionally report negative results without suppression (four failed resolution-selection strategies and the threshold sensitivity above), in line with best practice for computational benchmarking.

Declarations. This work is original, has not been published elsewhere and is not under consideration by another venue. All datasets are publicly available from GEO. Analysis code, processed matrices and all evaluation tables are provided as a reproducible pipeline. The author declares no competing interests.

We would be grateful for the committee's consideration and look forward to the reviewers' comments.

Sincerely,

Mingyu Sun